6. Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Android Spinner is a view similar to the dropdown list which is used to select one option from the list of options. It provides an easy way to select one item from the list of items and it shows a dropdown list of all values when we click on it. The default value of the android spinner will be the currently selected value and by using **Adapter** we can easily bind the items to the spinner objects. Generally, we populate our Spinner control with a list of items by using an **ArrayAdapter** in our Kotlin/Java file.

Project Name: AdapterDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <Spinner

       android:id="@+id/spinner"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"/>

   <ListView

       android:id="@+id/listView"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.adapterdemo;


import android.os.Bundle;

import android.view.View;

import android.widget.AdapterView;

import android.widget.ArrayAdapter;

import android.widget.ListView;

import android.widget.Spinner;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   Spinner spinner;

   ListView listView;


   String[] cities = {

           "Mumbai",

           "Pune",

           "Delhi",

           "Chennai",

           "Hyderabad"

   };
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        spinner = findViewById(R.id.spinner);
        listView = findViewById(R.id.listView);


        // Spinner Adapter
        ArrayAdapter<String> spinnerAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_dropdown_item,
                cities
        );


        spinner.setAdapter(spinnerAdapter);
        // ListView Adapter
        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                cities
        );
        listView.setAdapter(listAdapter);
        // Spinner Item Selection
        spinner.setOnItemSelectedListener(new
AdapterView.OnItemSelectedListener() {
```

```
            @Override
            public void onItemSelected(AdapterView<?> parent, View view,
                                       int position, long id) {


                Toast.makeText(
                        MainActivity.this,
                        "Selected: " + cities[position],
                        Toast.LENGTH_SHORT
                ).show();
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {


            }
        });
        // ListView Item Click
        listView.setOnItemClickListener((parent, view, position, id) -> {
            Toast.makeText(
                    MainActivity.this,
                    "Clicked: " + cities[position],
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```